CHAPTER 3

COMPONENTS OF THE COMPUTER SYSTEM

CHAPTER ESSENCE AT A GLANCE:
Master the physical and functional architecture of computers combining processor registers, hierarchical memory subsystems, multi-bus interconnections, and multi-layered system software.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Three-Bus Architecture Synergy:** The processor coordinates hardware peripherals through the unidirectional address bus, bidirectional data bus, and synchronizing control bus lines.

• **Hierarchical Memory Trade-Off:** Computer memory balances access speed, capacity, and cost per bit, cascading from high-speed CPU registers and cache down to volatile RAM and persistent secondary storage.

• **Volatile versus Non-Volatile Storage:** Dynamic RAM requires continuous electrical refresh cycles and loses content upon power disruption, whereas ROM and flash-based SSDs retain firmware instructions indefinitely.

• **Language Processing Hierarchy:** Compilers translate entire source code files into machine code ahead of execution, whereas interpreters parse and execute statements line-by-line in real time.

ESSENTIAL ARCHITECTURE & CORE SYNTAX CARDS

$$\text{Bus Bandwidth} = \text{Bus Width (bits)} \times \text{Clock Frequency (Hz)}$$

System Bus Data Transfer Rate Formula

$$\text{Speed: Registers} > \text{Cache} > \text{RAM} > \text{SSD} > \text{Hard Disk}$$

Hierarchical Memory Access Latency

$$\text{Addressable Locations} = 2^{\text{Address Bus Width}}$$

Maximum Memory Capacity Calculation

$$\text{Compiler: Whole File} \rightarrow \text{Object} \quad | \quad \text{Interpreter: Line-by-Line}$$

Language Translation Execution Models

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Differentiate clearly between SRAM (flip-flop based, ultra-fast, used for cache) and DRAM (capacitor based, needs continuous refresh, used for main memory).

• **Alert:** Remember that the address bus is strictly unidirectional (CPU to memory), whereas the data bus is bidirectional.

• **Alert:** Compilers report all syntax errors in batch mode; interpreters halt execution immediately upon encountering the first syntax error.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Distinguish between RAM and ROM on two key parameters. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Volatility:** RAM (Random Access Memory) is volatile; its stored data is lost immediately when power is turned off. ROM (Read Only Memory) is non-volatile; retains permanent firmware instructions indefinitely. **[1 Mark]**

• **Read-Write Operations:** RAM supports both high-speed read and write operations during execution. ROM is primarily read-only during normal computer operation. **[1 Mark]**

2. What is Cache Memory? Why is it placed between CPU and RAM? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Definition:** A small, high-speed static semiconductor memory that stores frequently accessed instructions and data. **[1 Mark]**

• **Role:** Bridging the speed mismatch between the ultra-fast CPU and relatively slower dynamic RAM, reducing CPU idle wait cycles. **[1 Mark]**

3. Explain the three types of system buses: Data Bus, Address Bus, and Control Bus. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Data Bus:** Bidirectional bus that transfers actual data words between the CPU, memory locations, and input-output peripheral controllers. **[1 Mark]**

• **Address Bus:** Unidirectional bus carrying physical memory addresses from the CPU to memory modules or I-O ports, determining maximum addressable memory. **[1 Mark]**

• **Control Bus:** Carries synchronization signals, read-write command lines, clock pulses, and interrupt signals from the Control Unit across all system components. **[1 Mark]**

4. Compare Compilers and Interpreters on three fundamental characteristics. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Translation Unit:** Compiler translates the entire source code file into machine object code at once; Interpreter translates and executes source code line-by-line. **[1 Mark]**

• **Execution Speed:** Compiled code executes significantly faster after build time; Interpreted code runs slower due to continuous runtime translation. **[1 Mark]**

• **Error Diagnosis:** Compiler generates a comprehensive list of all syntax errors after full scan; Interpreter halts execution at the first encountered error. **[1 Mark]**

5. Classify computer software into System Software and Application Software with appropriate subcategories. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **System Software:** Manages hardware resources and provides platform for applications; includes Operating Systems (Windows, Linux), Language Processors (Compilers, Assemblers), and Device Drivers. **[1.5 Marks]**

• **Application Software:** Designed to accomplish specific user tasks; includes General Purpose software (Word Processors, Spreadsheets) and Customized and Specific Purpose software (Payroll systems, School management). **[1.5 Marks]**

6. If a computer processor possesses a 32-bit address bus, calculate its maximum directly addressable primary memory capacity in gigabytes. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Formula):** Total addressable locations = 2 raised to the power of Address Bus Width. **[1 Mark]**

• **Step 2 (Calculation):** For 32-bit bus, total locations = 2^{32} bytes = 4,294,967,296 bytes. **[1 Mark]**

• **Step 3 (Gigabyte Conversion & Boxed Result):** 2^{32} bytes = $2^2 \times 2^{30}$ bytes = 4 × 1 GB = 4 GB. Boxed Result: Maximum addressable RAM capacity is 4 Gigabytes (4 GB). **[1 Mark]**

7. Explain secondary storage technologies: Magnetic Hard Disk, Optical Disc (DVD and Blu-ray), and Solid State Drive (SSD) with performance comparisons. **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• **Magnetic Hard Disk (HDD):** Uses spinning magnetic platters and mechanical read-write heads; high storage capacity at low cost, but vulnerable to mechanical shock and higher latency. **[1 Mark]**

• **Optical Discs (CD, DVD, Blu-ray):** Read and written using laser beams; portable and non-magnetic, but slower data transfer rates. **[1 Mark]**

• **Solid State Drive (SSD):** Utilizes non-volatile flash memory microchips with zero moving parts; delivers ultra-fast data read-write speeds, low power consumption, and high shock resistance. **[1 Mark]**